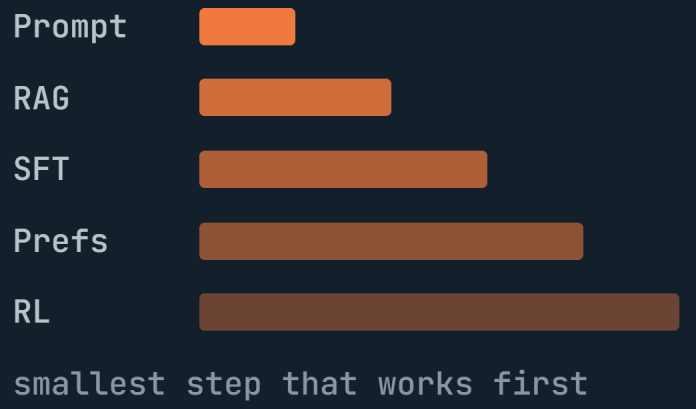
BOOK SUMMARY · 13 CHAPTERS · 4 PARTS

The Craft of Post-Training

A Practical Guide for AI Engineers and Developers



Chris von Csefalvay · No Starch Press, 2026

THE BOOK IN ONE SLIDE

Post-training gives a model its purpose

WHAT MATTERS, IN ORDER

Data quality > **Algorithm choice** > **Model size** > Compute

“Right of pre”

Everything after pre-training: fine-tuning, alignment, compression, evaluation, safety, deployment.

Smallest step first

Prompting → RAG → SFT → preference optimisation → RL.

Evaluate continuously

Layer your evaluation. Any single metric gets gamed (Goodhart’s law).

Data is the moat

Differentiation comes from proprietary data and expert judgment, not the “best” base model.

BOOK MAP

Thirteen chapters in four parts

PART I

Foundation

1. Post-training essentials
2. Before you fine-tune

PART II

Tools

3. Supervised fine-tuning
4. RL, RLHF and PPO
5. DPO, KTO, GRPO and friends
6. Evaluation

PART III

Craft

7. Quantization and compression
8. Domain adaptation
9. Agentic models
10. Reasoning

PART IV

Frontier

11. Synthetic data and self-play
12. Multimodal systems
13. Future directions

Parts I–II: whether to post-train, and the core tools. Parts III–IV: the craft of applying them, and the frontier.

Three scales of post-training

	Memscale	Flopscale (this book)	Powerscale
Who	Solo, hobbyists	Enterprise teams	Foundation labs
Compute	1 GPU (16–80 GB)	Tens to hundreds of GPUs	Thousands of GPUs
Cost per run	$10s–$100s	$1Ks–$10Ks	$1M+
Limiting factor	VRAM	ROI, unit economics	Energy, grid
Techniques	LoRA, QLoRA, gradient checkpointing	Full fine-tuning, RLHF, distillation	Multistage RLHF, synthetic data at scale

Techniques don't transfer automatically between scales. In the enterprise "crucial middle", the question is **"should we, and at what cost?"**

PART I · CHAPTER 1

Why prompting alone stalls in production

Foundation models are optimised for breadth and plausibility, not depth. Demos impress; deployments disappoint.

PROMPTING HITS THREE LIMITS

1 Context window size
2 Specificity
3 Consistency

→

PRODUCTION NEEDS THREE GUARANTEES

1 Consistency
2 Compliance
3 Domain-safe accuracy

PART I · CHAPTER 2

Use the smallest intervention that works

A knowledge problem calls for **RAG**. A behaviour problem calls for **fine-tuning**.

01 · START HERE

Prompting

Lowest maintenance. Exhaust it first.

02

RAG

Fresh or proprietary facts.

03

SFT

Behaviour you can show by example.

04

Preference optimisation

DPO, KTO, GRPO on pairs, ratings or verifiers.

05

RL

Qualities you can judge but can't write.

cost and maintenance rise →

Fine-tuning is an ongoing commitment

RULE OF THUMB

6–12 months

between full re-fine-tunes in fast-moving domains

WHERE THE ONGOING COST COMES FROM

- **New base model releases**
- **World and regulatory drift**
- **Maintaining training infrastructure**

Ask first: **“Can we maintain this indefinitely?”**

Four checks before picking a base model

Tokenizer

Frozen after pre-training. Test it on your own domain text first.

Size

70B with 1K examples ≈ 7B with 10K, but bigger costs more at every stage.

Open vs. closed

Open weights: control and privacy. APIs: easy start, but vendor lock-in.

Licence

“Open-weight” isn’t free for any use. Llama restricts products above 700M monthly active users.

Data is the number-one lever in SFT

CURATED

1,000

examples beat

~~40,000~~

MEDIOCRE

01 **Use the model's chat template.** A mismatch is one of the most common failure causes.

02 **Deduplicate.** Even 2–3 copies of an example cause memorisation.

03 **Read 20–50 outputs by hand.** Loss alone won't show you what's wrong.

04 **Checkpoint often.** The best checkpoint is rarely the last one.

SFT often solves ~80% of the problem in a week. Gains taper roughly logarithmically with more data.

Loss patterns point to likely causes

Pattern	Likely cause	Fix
Loss flat	LR too low, data bug, template mismatch	Raise LR 10×; check pipeline
Wild oscillation	LR too high	Lower LR 10×; add warm-up
Sudden spike	Gradient explosion	Gradient clipping
Train ↓, validation ↑	Overfitting	Early stopping; more data
Train ≈ 0	Memorisation	Fewer epochs; more diversity
Gradual rise	Catastrophic forgetting	Lower LR; regularise

The validation set must look like real deployment inputs. **Loss says something is wrong; outputs say what.**

PART II · CHAPTER 4

People judge better than they write

IF YOU CAN WRITE EXAMPLES

SFT

Formats, style, vocabulary, templates.

IF YOU CAN ONLY JUDGE

RL / preferences

Helpful, honest, harmless behaviour.

RLHF PIPELINE

Preferences → Reward model → Policy optimisation

Key numbers for an RLHF run

ANNOTATORS

> 0.7

Cohen’s kappa agreement. Below 0.4, fix the guidelines.

REWARD MODEL

> 70%

Held-out accuracy target (Bradley–Terry model).

PPO LR

5e-6

About 10× lower than SFT. Unstable? Lower it.

KL BETA

0.02

The most-tuned knob. Keeps the policy near its SFT reference.

Policy collapse

High reward, nonsense output → tighten KL, ensemble reward models

Length gaming

Longer answers farm reward → add a length penalty

Mode collapse

Outputs all look alike → add an entropy bonus

Debug order: reward scores → KL → read the outputs

Match the method to the data you have

	PPO	DPO	KTO	GRPO
Data	Reward model	Preference pairs	Unpaired up/down ratings	Reward model or verifier
Policy	On	Off	Off	On
Compute	Highest	Lowest	Low	Moderate
Infrastructure	Complex	SFT-like	SFT-like	Moderate
Best for	Max control and exploration	Simplicity	Existing feedback logs	Scale without a value network

Data quality matters more than algorithm choice. With equal implementation quality, online methods beat offline ones.

Layer evaluation from seconds to weeks

Level	Speed	Cost	Use for
Automatic metrics	Seconds	~0	Every checkpoint; regressions
LLM-as-judge	Minutes	Moderate	Filtering candidates; analysis
Human evaluation	Hours–days	High	Deployment decisions; calibration
A/B testing	Weeks	Variable	Production truth



Metric divergence is the key signal.

Goodhart’s law: any single metric gets gamed. Evaluation continues after deployment.

LLM judges carry four biases

Position bias

Fix: swap the order; keep only verdicts that agree.

Length bias

Fix: say brevity is valued, or score conciseness separately.

Self-preference

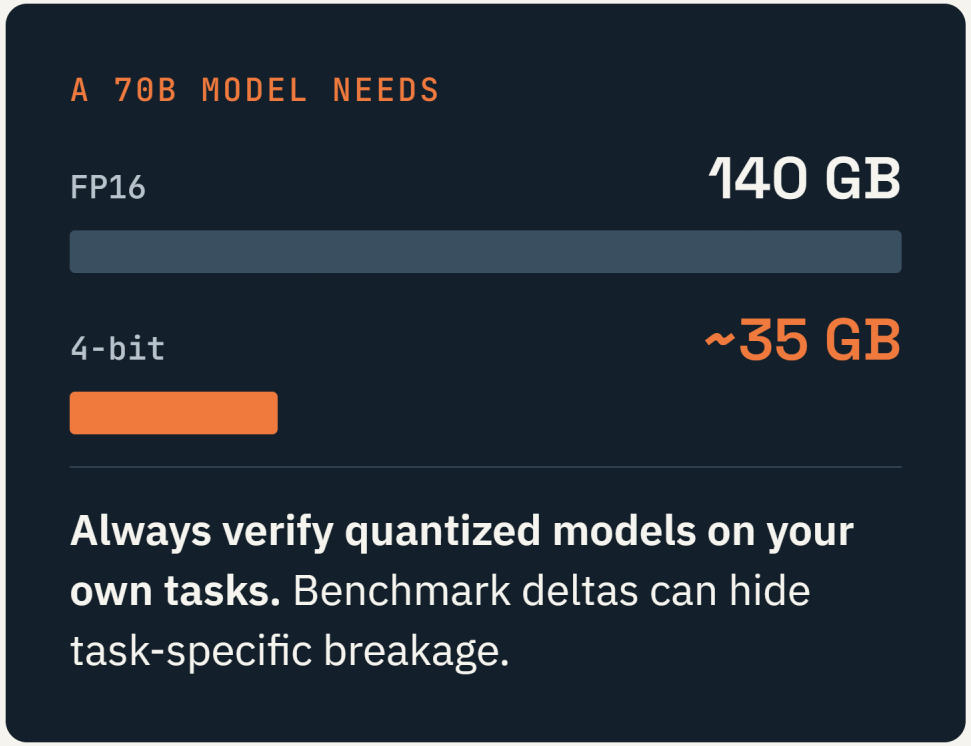
Fix: use a judge from a different model family.

Confidence bias

Fix: add calibration examples that reward appropriate hedging.

Recalibrate judges against human ratings periodically.

Compression makes models deployable



If you need	Default
Accurate 4-bit	AWQ (<1% loss)
Local inference	GGUF Q4_K_M; Q5_K_M for balance
Fine-tuning, low VRAM	QLoRA (30B on 24 GB)
Efficient tuning	LoRA (0.1–1% of params)
Real pruning speedups	Structured or 2:4 sparsity
Fast serving	vLLM, continuous batching, speculative decoding

Domain adaptation is a ladder too

Strategy	Data	Cost	When to use
Prompting	None	Negligible	Well covered in pre-training
RAG	Documents	Low–moderate	Current or proprietary facts
Domain SFT	100s–1,000s examples	Moderate	Task behaviour
Continued pre-training	Billions of tokens	High	Severe mismatch; new terms
Domain RL	Expert preferences	Very high	Quality beyond SFT
Full pipeline	All of the above	Highest	Regulated, production-critical

Avoid forgetting: replay general data, keep the learning rate low, monitor general skills. **Plan for decay:** rebase onto new models and refresh data regularly.

Tool use is learned; tools will fail

1. **Decide** whether a tool is needed →
2. **Choose** the right tool →
3. **Construct** the call and arguments →
4. **Interpret** the result →
5. **Integrate** it into the response

TRAIN ON FAILURE TRACES SO IT DOESN'T

- Ignore errors
- Invent results
- Retry in a loop
- Give up too early

SAFETY

Sandboxing and least privilege
Trained refusal of harmful actions
Humans in the loop for irreversible or high-stakes actions

Evaluate outcomes over actions: task success first, then efficiency

Reward the process; verify what you can

AT 95% PER-STEP ACCURACY

60%

of 10-step chains are valid

36%

of 20-step chains are valid

	Outcome reward (ORM)	Process reward (PRM)
Signal	Final answer only	Every step
Labels	Cheap	Expensive; needs experts
Diagnosis	Unclear where it failed	Pinpoints the failing step
Best for	Simple tasks, lots of data	Complex, verifiable reasoning

Verifiable rewards (RLVR) scale without annotators. Proof assistants (Lean 4, Coq) and code tests act as ground truth.

PART IV · CHAPTER 11

Scale synthetically, anchor with humans

The only test that matters: **does the data improve the model?**

GENERATE

Simple to advanced

Prompting, critique-and-revise, Self-Instruct, Evol-Instruct, persona-driven and textbook-style generation.

FILTER

Less can be more

The best half of a dataset can beat the whole. Score quality, deduplicate, enforce diversity.

SELF-PLAY

Plateaus without an anchor

Debate, SPIN, STaR and iterated distillation drift without external grounding.

⚠ **Model collapse:** diversity shrinks across generations. Keep a human-data reservoir as the anchor.

Same principles, double the annotation

ViT
Vision encoder

→

MATTERS MOST
Projection layer

→

LANGUAGE
LLM

WATCH FOR
Visual hallucination, which the image makes feel grounded. Annotation costs 2–3× text-only.

What to train	When	Data	Forgetting risk
Projection only	Default	1K–10K	Low
LoRA on LM + projection	Balanced	5K–50K	Low–moderate
LM + projection	Deeper adaptation	10K–100K	Moderate
Vision encoder + projection	Novel visual domain	100K+	High

Default base: Qwen2.5-VL 7B (Apache 2.0, strong document OCR)

PART IV · CHAPTER 13

Invest where the rising tide won’t lift

Proprietary data

Internal processes

Domain expertise

Generic skills like summarisation and translation get commoditised. Buy where vendor offerings suffice; build where you differentiate.

Talent is the real bottleneck

more than compute or data

TAKEAWAYS

Ten principles that outlast the techniques

01 **Minimal intervention**
Prompt → RAG → SFT → preferences → RL.

02 **Data quality beats everything**
You can't out-algorithm bad data.

03 **SFT before RL, always**
RL without SFT wastes GPU time.

04 **Recognise vs. demonstrate**
Write examples: SFT. Only judge: RL or preferences.

05 **Read the outputs**
Loss says something's wrong; outputs say what.

06 **Mind Goodhart's law**
Measure from several angles; divergence is signal.

07 **Stay close to the reference**
KL penalties, low learning rates, replay data.

08 **Verifiable domains scale**
Code tests and proof checkers replace labels.

09 **Humans as anchor**
For synthetic data, AI judges and evaluation.

10 **Plan for maintenance**
Budget to retune every 6–12 months.

TAKEAWAYS

Pick the tool from what you have

If you have or need...	Use
Fresh or proprietary facts	RAG
Consistent format, style or voice	SFT (LoRA)
Limited GPU memory	QLoRA
Pairwise preference data	DPO
Thumbs-up/down user feedback	KTO
Verifiable rewards or a strong reward model, at scale	GRPO
Maximum control or exploration	PPO

If you have or need...	Use
SFT and alignment in one step	ORPO
Severe domain mismatch	Continued pre-training + replay
A big model you can’t deploy	Distillation or quantization
Fast local inference	GGUF Q4_K_M–Q5_K_M
Multistep reasoning quality	PRM + search, or RLVR
More data, cheaply	Synthetic + filtering + human anchor

WHAT LASTS

Data quality > algorithm choice > model size > compute.

Companion code: github.com/chrisvoncsefalvay/craft-of-post-training